	Social Science/ Humanities/Fine Arts Elective	3	

* BS bound students are required to take CIVE400 - Summative Learning (BS) Project instead.
Note: Engineering Technical Electives (levels 400 or above) are selected as such:
6 credit restricted Departmental Electives
9 credits from any Engineering discipline at the 400 level or above.

Course Coding

Code numbers designate the courses offered in the Civil Engineering program in the form of (CIVE abc), where:

a	Year (level): 2 = first year, 3 = second Year, 4 = Third year, 5 = Fourth Year
b	Concentration Areas (as follows) 0: Common Courses and Construction Management; 1: Structural Analysis and Design; 2: Construction Materials and Geotechnics; 4: Public Works; 7: Water Resources and Treatment;
c	Course sequence in area: 0, 1, …, 9

Courses Description

I. Mandatory Courses

Non-Major Courses

A description of the non-major mandatory courses follows.

CHEM 211	**Environmental Chemistry and Lab**	**3(2,2)**
This course provides an opportunity to develop an understanding of several basic environmental functions, the complicated nature of environmental systems, and the need for multidisciplinary solutions to environmental problems. Topics covered include the hydrosphere, water quantity and quality, soil and the soil ecosystem, biological resources, waste disposal, air pollution, ozone depletion, acid rain, and global warming. The course also includes a set of experiments that provide students with practical experience in various environmental analysis settings, including air and water quality.		

COSC 214	**Introduction to Programming**	**3(2,2)**
This course presents the fundamentals of structured and modular programming concepts. It covers primitive data types, expressions, control statements, functions, arrays, basic search and sorting algorithms, and an introduction to pointers. **Prerequisite**: None. Equivalent to CCEE 214.		

ELEC 210	**Electrical Circuits**	**3(3,0)**
Circuit elements and laws; mesh and node equations; network theorems; energy-storage elements; RC, RL, and RLC circuits; Laplace Transform; sinusoids, phasors,		

and an introduction to network theory. Sinusoidal (AC) steady-state, mutual inductance, transformers, and an introduction to three-phase circuits. **Prerequisite**: PHYS 191 or Equivalent

ENGL 210	**English Composition and Rhetoric**	**3(3,0)**
This course reviews the fundamentals of good academic writing in English, teaches essay writing essentials and research skills in two rhetorical modes (persuasion and argumentation), and provides practice in writing essays in these modes, research papers, and oral presentations. **Prerequisite:** ENGL 101 or TOEFL 550+ (paper) or 80+ (computer).		

ENGL 217	**Professional English Communication**	**3(3,0)**
This required course is designed to help students develop effective professional communication skills, both orally and in writing. In this course, students learn how to write emails, memos, letters, proposals, reports, and other forms of employment correspondence. In addition, this course helps students sharpen their presentation skills. Broadly, this course enables students to behave professionally and effectively in their prospective jobs. **Prerequisite**: ENGL 210.		

ENGR 300	**Engineering Economics and Management**	**3(3,0)**
Concepts and techniques in basic Engineering economy principles and applications. Interest and financial mathematics: present worth, annual worth, benefit/cost ratio, internal rate of return, multiple alternatives, income tax, inflation, Risk analysis, Investment and investment choice, equivalence, loans, cost of capital, retirement and replacement, sector analysis and viewpoint, sensitivity analysis, accounting, and financial statements. **Prerequisite**: None. Annually.		

ENGR 510	**Engineering Project Management**	**3(3,0)**
The course covers key components of engineering project management, including projects election and planning, project time management, cost estimation and pricing, contract and specifications, quality management, engineering ethics and professional conduct, realizing the impact of engineering solutions in various contexts (global, economic, environmental, societal, etc...), sustainability in engineering designs, human resources consideration, communications, risk management, and procurement management. **Prerequisite:** ENGR 300.		

MATH 215	**Calculus III**	**3(3,0)**
This course covers complex numbers, trigonometric functions, inverse trigonometric functions, hyperbolic functions, and integration techniques. The course also covers infinite sequences and series: limits of sequences of numbers, bounded sequences, integral test for series, comparison tests, ratio and root tests, alternating series, absolute and conditional convergence, power series, Taylor and MacLaurin series, and applications of power series. Polar functions, polar coordinates, and graphing of polar curves are also covered. In addition, topics from multivariable calculus are introduced: double integrals, applications to double integrals, and double integrals in polar form.		

MATH 216	Calculus IV	3(3,0)
Multivariable calculus and vector analysis for three-dimensional problems. Topics include 3-D vectors, dot and cross products; quadratic surfaces; limits, continuity, and partial derivatives; gradient, chain rule, and directional derivatives; tangent planes, differentials, extrema, and saddle points; triple integrals in rectangular, cylindrical, and spherical coordinates with applications to mass and moments; vector fields, work, flux, and Green's Theorem; and an introduction to Fourier series. **Prerequisite:** MATH 215.		

MATH 311	Linear Algebra with Applications	3(3,0)
Systems of linear equations, matrix algebra, linear transformations, determinants, vector spaces, eigenvalues and eigenvectors, symmetric matrices, orthogonality, diagonalization.		

MATH 314	Ordinary Differential Equations	3(3,0)
First-order linear differential equations; linear differential equations of second and higher order; linear differential equations with variable coefficients; series solutions; systems of differential equations; Laplace transforms and their inverses. **Prerequisite:** MATH 216.		

MATH 351	Probability and Statistics	3(3,0)
Probability and conditional probability; discrete and continuous random variables; marginal distributions; expectation, variance, mean, median, covariance, and correlation; conditional expectation; Normal distribution; sampling distribution; prediction and confidence intervals; hypothesis testing; and regression line and correlation coefficients. **Prerequisite**: MATH 215		

MATH 421	Numerical Analysis	3(3,0)
Error Analysis, solutions of nonlinear equations using fixed-point, Newton-Raphson-Muller's methods, solution of linear systems using Gaussian elimination-iterative methods, interpolation and approximation using Taylor series-Lagrange approximation-Newton polynomials, numerical differentiation and integration, numerical optimization, solutions of ordinary and partial differential equations using Euler's and Heun's and Rung-Kutta methods, MATLAB codes related to the above-mentioned topics. **Prerequisite**: MATH 311.		

MECH 220	Dynamics	3(3,0)
Brief review of statics; proper use of different coordinate systems: Cartesian and normal/tangential axes; kinematics of a particle: rectilinear, curvilinear, and relative motion; kinetics of a particle: force and acceleration, work and energy, impulse and momentum; planar kinematics of a rigid body: translating and rotating axes, absolute and relative motion; planar kinetics of a rigid body: force and acceleration, work and energy, impulse and momentum. **Prerequisite**: CIVE 211. Annually.		

MECH 320	Mechanics of Materials	3(3,0)
Mechanical stress, strain, and stress/strain relations under pure tension, compression, pure bending, and pure torsion of circular bars; 1-D thermal strain & stress; combined		

stresses; principal stresses; plane stress transformation & Mohr's circle; basic 3-D elastic stress/strain relations; Euler-elastic buckling of columns. **Prerequisite**: CIVE 211. Annually.

MECH 333	**Thermal Fluid Sciences**	**3(3,0)**
This course seeks to impart fundamental concepts of thermal-fluid (including thermodynamics, fluid mechanics, and heat transfer) to non-mechanical engineering students. These cover: thermodynamic state and properties of pure substances; concepts of work and heat; energy analysis of closed systems (First Law) and simple flow devices; fluid statics; Bernoulli's equation; Conservation of mass, momentum & energy; viscous flow in pipes; introduction to conduction, convection, and radiation heat transfer. **Prerequisite**: MATH 216. Annually.		

PHYS 210	**Fundamentals of Physics and Lab**	**3(3,1)**
Welcome to the Introductory Physics course. This course consists of: Review of Classical Mechanics; Fluid Mechanics; General Properties of Waves; Electric Forces; Electric Fields for Discrete and Continuous Charge Distributions; Gauss's Law; Electric Potential; Kirchhoff's Rules; Magnetic Fields and Forces; and Qualitative Discussion of Maxwell's Equations. **Prerequisite**: None.		

Major Courses

A description of the civil engineering mandatory courses follows.

CIVE 200	**Introduction to Civil Engineering**	**2(2,0)**
A broad introductory course that exposes students to the history and heritage of civil engineering; specialized sub-disciplines (Structural, Transportation, Construction Materials, Geotechnical, Environmental); challenges of the civil engineering profession; professionalism, professional registration, and societies; and Codes and Standards. This course also introduces students to the business aspects of the civil engineering profession, including construction management and engineering economics.		

CIVE 202	**Civil and Architectural Drawings**	**3(2,2)**
Essential engineering drawing concepts for civil and architectural applications, along with computer drafting skills in 2D and 3D, are taught and applied to produce floor plans, sections, elevations, and reinforcement detailing for reinforced concrete elements. Emphasis is directed at teaching students proper placement of symbols, annotation, and scheduling information required for site construction. **Prerequisite**: None. Annually.		

CIVE 211	**Statics**	**3(3,0)**
Static laws; force vectors and operations; force systems and moments; free-body diagrams; equilibrium of particles and rigid bodies in 2D and 3D; plane trusses and frames; cables; internal forces: shear and moment diagrams; centroid and center of mass; moment of inertia; and friction. **Prerequisite**: None. Annually.		

CIVE 240	Surveying	3(2,2)
Surveying fundamentals; theory of errors in observations; leveling; distance measurement; angles, azimuths, and bearings; traversing; topographic surveys and maps. The course also has a field component in which students work in groups to conduct surveys on the topics covered and write technical reports. **Prerequisite**: None. Annually.		

CIVE 312	Structural Analysis I	3(3,0)
Shear and bending moment diagrams for frames, equilibrium, stability, and determinacy; influence lines for determinate structures; Elastic deformation of beams, frames, and trusses; introduction to indeterminate structures. **Prerequisite**: MECH 320. Annually.		

CIVE 314	Concrete I	3(3,0)
Behavior of reinforced concrete structural elements; concepts of design and proportioning sections for strength and serviceability; theory of flexure and shear; analysis and design of beams for flexure and shear; analysis for crack width and deflection; design philosophies and process; design of one way solid and joist slabs; bond theory; development, anchorage and splicing of reinforcement; bars cut-off; continuous beams and one-way slabs. **Prerequisite**: MECH 320. Annually.		

CIVE 321	Construction Materials	3(3,0)
Properties and tests of materials used in the construction of civil structures, such as aggregate, cement, mixing water, admixtures, steel, masonry, Portland cement concrete, asphalt concrete, and timber; laboratory and field measurement techniques to assess material properties and performance; emphasis on fresh and hardened concrete, types of concrete and applications, and concrete mix design. Annually.		

CIVE 321L	Construction Materials Laboratory	1(0,2)
Los Angeles abrasion test; specific gravity and absorption of coarse and fine aggregates and soils; moisture content; bulk unit weight and voids in aggregates and soils; sieve analysis; concrete mix design; mixing concrete and slump test; compressive strength of concrete; absorption, moisture content and density of concrete masonry units. Report writing and data analysis. **Co-req**: CIVE 321. Annually.		

CIVE 341	Highway Engineering	3(3,0)
Introduction to highway engineering standards; types and classifications of roads; characteristics of the driver, the pedestrian, the vehicle, and the road; highway location and survey methods; earthwork, geometric design of highway components: vertical and horizontal alignment, transition curves, superelevations and intersections, highway materials and evaluations, construction techniques and plants, quality control and testing, visual assessment, and maintenance. **Prerequisite**: CIVE 240. Annually.		

CIVE 342	Traffic Engineering	3(3,0)
Social and economic impact of traffic engineering on society; basis of traffic engineering; transportation planning; traffic flow theory; traffic studies and analysis; travel demand modeling and forecasting; queuing theory; capacity analysis; level of service; traffic control and analysis at signalized intersections. **Prerequisite**: CIVE 341. Annually.		

CIVE 400	Summative Learning (BS) Project	3(3,0)
Team-oriented, project-based experience in which students work on a meaningful problem of concern to industry or the community at large, or to develop a promising idea. The project involves whatever is needed to achieve the intended outcome: literature review, design work, data collection, experimentation, data analysis, report writing, and presentation. **Prerequisite**: ENGL 217 and Senior Standing. Annually.		

CIVE 413	Computer Modeling of Structures	3(3,0)
Principles of structural modeling of different types of structural elements: trusses, beams, columns, and frames; modeling the structural behavior of buildings under the effect of vertical and lateral loads; computational applications using structural analysis software; case studies and projects. **Prerequisite**: CIVE 312. Annually.		

CIVE 415	Concrete II	3(3,0)
Short columns, slender columns, uniaxially and biaxially bent columns; two-way slab behavior, analysis, and design; retaining walls; wall footings, concentrically and eccentrically loaded single column footings, and combined footings; staircases. **Prerequisite**: CIVE 312 and CIVE 314. Annually.		

CIVE 417	Steel Structures	3(3,0)
Structural steel properties, loads, design methods, analysis, and design of members and connections in steel structures: tension, compression, and flexure members. **Prerequisite**: CIVE 312. Annually.		

CIVE 423	Soil Mechanics	3(3,0)
Soil classification; origin of soil and grain size; soil structure, weight-volume relationships; plasticity and structure of soil; soil compaction; permeability; seepage, stresses in a soil mass; compressibility of soil and consolidation; shear strength of soil. Prerequisites: MECH 320 and GEOL 221. Annually.		

CIVE 423L	Soil Mechanics Laboratory	1(0,2)
Experimental testing of soil: Sieve Analysis, Atterberg Limits, Proctor test; sand cone field density measurement, consolidation, and direct shear test; report writing, and data analysis. Co-requisite: CIVE 423. Annually.		

CIVE 424	Foundation Engineering	3(3,0)
Geotechnical site investigation; field and laboratory tests; characterization of strata; foundation systems; ultimate bearing capacity and settlement of shallow foundations (spread footings, strip footings, combined footings, and mat); effect of water table level		

on bearing capacity; structural design of mat foundations; ultimate bearing capacity and settlement of deep foundations (single pile and group of piles); pile load test. **Prerequisite**: CIVE 423. Annually.

CIVE 432	**Environmental Engineering**	**3(3,0)**
Chemical and biological processes related to environmental science and engineering, materials and energy balances; environmental systems analysis with application to air, water, and soil; hydrology; water treatment; water quality management; wastewater treatment; air pollution; noise pollution; solid waste management; analysis of environmental problems and development of engineering solutions. **Prerequisite**: CHEM 211. Annually.		

CIVE 435	**Hydraulics**	**3(3,0)**
Engineering approaches to the measurement, control, and conveyance of water flows, with particular emphasis on the analysis, design, characteristics, and selection of hydraulic models, as well as the design of water and wastewater systems. **Prerequisite**: MECH 333. Annually.		

CIVE 499	**Co-op Work Experience**	**3(3, 0)**
This Co-op work experience is designed to provide students with an eight-week short-term opportunity in Civil and Environmental Engineering. Students will be aligned with discipline-related industries, institutions of higher learning, or government agencies. **Prerequisite**s: ENGL 217; Senior Standing.		

CIVE 501	**Construction Planning and Management**	**3(3,0)**
Fundamental elements of management of civil engineering projects; roles of all participants in the process: owners, designers, contractors, and suppliers; emphasis on contractual aspect, project estimate, planning and control of construction projects; management functions, network techniques (CPM), resource scheduling, construction financing, and cost/schedule relationship. **Prerequisite**: Senior standing. Annually.		

CIVE 507	**Structural Analysis II**	**3(3,0)**
Statically indeterminate structures, force method (beams, trusses, and frames). Cable-supported indeterminate structures. Temperature and settlement effects. Displacement methods: slope-deflection equations and moment distribution (beams and rigid frames). Qualitative influence lines for statically indeterminate beams. **Prerequisite**: CIVE 312. Annually.		

CIVE 508	AI Applications in Civil Engineering	**3(3,0)**
This course introduces students to the fundamentals of Artificial Intelligence (AI) and its practical applications in civil engineering. The course includes hands-on programming, case studies from real-world civil projects, and AI-enhanced tools to empower students with the skills to lead data- and AI-driven innovation in infrastructure and construction. **Prerequisite**: CCEE 214. Annually.		